



National University of Computer and Emerging Sciences, Karachi
Spring-2023, FAST School of Computing
Final Examination



23rd May 2023, 08:30 am – 11:30 am

Course Code: CS2005	Course Name: Database Systems
Instructor Name: Anam Qureshi	
Student Roll No:	Section:

Instructions:

- Return the question paper. Don't write anything on the question paper, except your Roll # & Section #.
- Read each question completely before answering it. There are **7 questions and 3 pages**.
- In case of any ambiguity, you may make assumptions. But your assumptions should not contradict any statement in the question paper.
- All the answers must be solved according to the sequence given in the question paper.
- This paper is subjective. Write the answers only on the answer sheet.

Time: 180 minutes.

Max Marks:50

Question#01 [CLO 2]

Marks: 8

- A. Highlight the differences between basic, conservative, strict, and rigorous two-phase locking protocols.
- B. Write down a schedule that must comprise three transactions i.e., T1, T2, and T3 with different read and write operations. Convert the schedule into a rigorous two-phase locking schedule.

Question #02 [CLO 2]

Marks: 6

Explain the levels of isolation in a transaction and specify which isolation level resolves each concurrency problem, such as lost update, temporary update, and incorrect summary problems.

Question #03 [CLO 2]

Marks: 8

Write down an algorithm for testing the serializability of a schedule. Also, give an example of a conflict serializable schedule.

Question #04 [CLO 3]

Marks: 7.5

Pakwheels car company manages the production and sales of various car models. The company keeps track of information such as car models, production quantities, sales records, and customer details. Here are some details of the scenario:

- The company has a table named "CarModels" that stores information about different car models, including their unique model IDs, names, and prices.
- The company maintains a table named "Production" to track the production details of each car model. The table contains the model ID, production date, and the quantity of cars produced on that date.

- The company has a table called “Sales” to record the sales transactions. It includes information such as the sales ID, customer ID, car model ID, date of sale, and the quantity of cars sold.
- The company keeps a table name “Customers” to store customer information. It contains details like customer ID, name, address, and contact information.

Write down the relational algebraic expressions for the following:

- Retrieve the name and prices of all car models.
- Find the total quantity of cars produced for a specific model.
- Determine the total revenue generated from car sales.
- Find the car models that have not been sold yet.
- Determine the average price of the sold cars.

Question #05 [CLO 3]

Marks: 7.5

Consider the scenario of Pakwheels (mentioned in Question #05). Write down the SQL statements for the following:

- Retrieve the details of all the customers who have made a purchase.
- Find the car models that have been sold more than 20 times
- Retrieve the customer details who have made purchases above the average purchase amount
- Find the car models with the highest and lowest sales quantity
- Retrieve the top 5 customers with the highest total purchase amount.

Question #06 [CLO 3]

Marks: 6.5

Computer Zone manages their customer’s orders. Their system has a single table called “OrderDetails” to store information about each order as shown in Table 01. They are facing data redundancy and anomalies in the current table design. As a database administrator, you have been tasked with normalizing the “OrderDetails” table up to the third normal form (3NF).

Table 01: Order Details of Computer Zone Store

Order ID	Customer ID	Customer Name	Customer Address	Product ID	Product Name	Product Price	Quantity	Order Date	Shipping Address
1	101	Ali	Karachi	201	Laptop	125K	2	18-5-20203	Karachi
2	102	Ahmed	Lahore	202	RAM	15K	1	19-5-2023	Lahore
3	103	Alyan	Quetta	203	Head-phones	10K	3	20-5-2023	Quetta

Question #07: [CLO 1]**Marks: 6.5**

Suppose you are designing a database system for a video game company. The company develops and publishes various video games across different platforms. Players can create accounts, purchase games, join multiplayer sessions, and communicate with other players. The company also keeps track of game reviews and ratings. Design an Entity-Relationship Diagram (ERD) to represent the following requirement specifications of the given scenario:

The system is comprised of a Game with its attributes as game_id, title, genre, platform, and release date. The Player information consists of player_id, username, email, password, and registration date. The Purchase details have purchase_id, player_id, and game_id. The Session information includes session_id, game_id, player_id, start_time, and end_time. The Reviews are represented as review_id, game_id, player_id, rating, and comments.

The relationship among the above entities can be defined as follows:

- A game can have multiple purchases and multiple reviews
- A player can make multiple purchases, write multiple reviews, and participate in multiple sessions.
- A purchase is made by a single player and can be associated with a single game.
- A session is associated with a single game and a single player.
- A review is written by a single player for a single game.

Good Luck!